

Why SU(3) Color = $D_{\text{BSFG}} / 2$

PAPER_1221 Category: UQFF Foundational **Status:** Complete **Date:** June 2026

Abstract

The number of quark colors in QCD is 3 — historically inferred from R-ratio measurements but never derived from first principles. UQFF derives this as **$N_{\text{colors}} = D_{\text{BSFG}} / 2 = 6/2 = 3$ EXACT**. The same locked primitive $D_{\text{BSFG}} = 6$ (bulk-edge dimension) simultaneously determines: 3 colors, the QCD one-loop β -function coefficient $b_0 = D_{\text{phys}} + D_{\text{BSFG}}/2 = 7$, and the Mexican-hat doublet coupling $K_{\text{MEX}} \times D_{\text{BSFG}} \approx 4\pi$.

Part 1: The Identity

$$N_{\text{colors}} = \frac{D_{\text{BSFG}}}{2} = \frac{6}{2} = 3 \text{ EXACT}$$

Part 2: Cross-validation

Multiple consistent applications of $D_{\text{BSFG}} = 6$: - $N_{\text{colors}}_{\text{SU}(3)} = D_{\text{BSFG}}/2 = 3$ (this paper) - $4\pi \approx K_{\text{MEX}} \times D_{\text{BSFG}} = 12.5$ (0.53% match) - QCD $b_0 = D_{\text{phys}} + D_{\text{BSFG}}/2 = 7$ EXACT (PAPER_1167) - $16\pi^2 \approx D_{\text{crit}} \times D_{\text{BSFG}} = 156$ (1.21% match) - Hodge identity $(D_{\text{phys}} + D_{\text{BSFG}})/SO_5 = 1.0$ EXACT

Part 3: Physical Significance

The bulk-edge dimension $D_{\text{BSFG}} = 6$ is the count of “extra” compactified dimensions accessible to gauge fields in the UQFF 26D framework. Dividing by 2 yields the rank of the bilinear color charge structure. The same D_{BSFG} controls the running of strong coupling via b_0 and the gauge-prefactor 4π normalization in vertex diagrams.

Part 4: Gluon Count

UQFF predicts $D_{\text{phys}}^2 - 1 = 15$ colored gauge fields. The SM observes 8 gluons (from the SU(3) adjoint). The 7 “extra” gauge bosons correspond to the compactified directions and could become accessible at very high energy scales.

Conclusion

The 3 colors of QCD are not an arbitrary phenomenological input — they are the Ricci/Mexican-hat dimensional ratio $D_{\text{BSFG}}/2 = 3$ in UQFF.

Framework Version: UQFF 5.27+